School & Madrasa Management SaaS — Full Project Plan (v2 — merged with competitive analysis)

> **Handoff Document for Claude Code** Complete architecture, dashboard design, permission model, database schema, and implementation phases for a multi-tenant school/madrasa management SaaS targeting Bangladesh.
>
> **v2 updates:** merged all findings from `FEATURE_COMPARISON.md` (Bornomala audit). New tables, default seeds, madrasa module, multi-branch, and 10 Tier-1 differentiators all folded in.
>
> **Stack:** Next.js 14+ (App Router) + TypeScript + Supabase (PostgreSQL, Auth, Storage, Realtime, Edge Functions) + Tailwind CSS + shadcn/ui
>
> **Primary language:** Bangla (bn) with English (en) and Arabic (ar, RTL) fallback. Unicode-first (no legacy Bijoy/SutonnyMJ).

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1. Product Vision

A multi-tenant SaaS that lets any school or madrasa in Bangladesh sign up, configure their institution, and manage every aspect of operation — academic, financial, HR, communication, and community. Four audiences, four dashboards, one platform.

Positioning: "Bornomala has — so do we. Plus: online payments, parent mobile app, live dashboard, madrasa-specific tools, offline mode, multi-branch support, AI insights. Same or lower price."

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2. Four Dashboard Types

Routes are grouped so middleware enforces access cleanly.

2.1 Super Admin — `/super-admin/*`

Platform owner. Manages tenants, subscriptions, billing, feature flags, SMS credits, platform analytics, global announcements, plan CRUD, system health, audit trails.

2.2 Principal / School Admin — `/school/[schoolSlug]/admin/*`

Principal, Vice-Principal, School Owner.

Scope: full school control, all finances visible including profit/loss, staff salary, branch comparison (if multi-branch).

Capabilities: school profile, branches, staff CRUD, granular permission grants, admission inquiry pipeline, students (single + bulk + shift + ledger), classes/sections/subjects, class-subject + teacher-subject assignment, fee structure, fee heads, expense heads, donation campaigns, investments, exams (routine → admit card → seating → marks → marksheet → result publish), certificates, notices, SMS/WhatsApp/Push composer, reports (20+ types), settings.

2.3 Teacher — `/school/[schoolSlug]/teacher/*`

Class Teacher (মূল শ্রেণিশিক্ষক): full control over assigned class/section. **Subject Teacher (বিষয় শিক্ষক):** only assigned subject × class combos. **Madrasa Ustadh:** Sabaq-Sabqi-Manzil daily entry, Hifz progress entry, Kitab reading log.

Never sees: finance, salary, other classes/subjects unless explicitly granted.

2.4 Parent / Student Portal — `/school/[schoolSlug]/portal/*`

Mobile-first PWA. Parent sees child's attendance, results, fees, homework, notices, messages. Pays online (bKash/Nagad/Rocket/Card). Student sees read-only subset + homework submission + attendance streaks. Multi-child support for siblings.

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3. Permission System (RBAC + Scoped)

3.1 Roles

SUPER_ADMIN

SCHOOL_ADMIN (Principal)

VICE_PRINCIPAL

ACCOUNTANT

BRANCH_ADMIN (multi-branch)

CLASS_TEACHER

SUBJECT_TEACHER

MADRASA_USTADH (Hifz / Kitab / Sabaq instructor)

LIBRARIAN

TRANSPORT_MANAGER

HOSTEL_WARDEN

CANTEEN_MANAGER (v2)

COUNSELOR (v2)

STUDENT

PARENT

3.2 Permission Tuple

`(action, resource, scope)`

- **action:** view, create, update, delete, approve, export
- **resource:** student, attendance, marks, fee, exam, salary, class, subject, notice, report_card, finance_report, donation, investment, hifz, sabaq, certificate, etc.
- **scope:** school, branch:{id}, class:{id}, section:{id}, subject:{id}, student:{id}, self

3.3 Enforcement — three layers

1. **DB:** Supabase RLS on every tenant table using `auth.uid()` + `user_permissions`.
2. **API:** central `requirePermission(user, action, resource, scope)` helper at top of every server action.
3. **UI:** `usePermission()` hook for conditional render.

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4. Multi-Tenancy + Multi-Branch

Tenancy: Shared database, shared schema, `school_id` discriminator + RLS. Composite indexes start with `school_id`.

Multi-branch (NEW): A school can have N branches (`school_branches`). Every tenant-owned row has optional `branch_id`. Principal sees all branches; Branch Admin sees only assigned branch. Consolidated reporting across branches for Principal.

Tenant identification: Path-based `/school/[slug]/...` for v1. Custom domain (`schoolname.com`) for Enterprise tier v2.

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5. Database Schema (Core Tables)

5.1 Tenant, Branches, Users

```
schools (

  id, slug, name_bn, name_en, eiin, address, logo_url,

  type,                          -- 'school' | 'madrasa' | 'both'

  subscription_plan_id, subscription_status, trial_ends_at,

  settings JSONB, created_at

)

school_branches (

  id, school_id, name, address, is_primary, settings JSONB

)

school_users (

  id, school_id, branch_id NULLABLE, user_id, role,

  employee_code, joined_at, status,

  UNIQUE(school_id, user_id, branch_id)

)

user_permissions (
```

```
  id, school_user_id, action, resource,
  scope_type, scope_id,
  granted_by, granted_at, expires_at
)
audit_logs (
  id, school_id, user_id, action, resource_type, resource_id,
  meta JSONB, ip, user_agent, created_at
)
```

5.2 Academic

```
academic_years (id, school_id, name, start_date, end_date, is_active)
classes (id, school_id, branch_id, name_bn, name_en, display_order, stream)
  -- stream: 'general' | 'hifz' | 'kitab' | 'nazera' | 'science' | 'commerce' | 'arts'
sections (id, class_id, name, capacity, class_teacher_id)
subjects (id, school_id, class_id, name_bn, name_en, name_ar,
  code, full_marks, pass_marks, is_optional)
class_subject_assignments (id, class_id, subject_id, academic_year_id)
teacher_assignments (
  id, school_user_id, section_id, subject_id,   -- subject_id NULL = class teacher
  academic_year_id, role_type                   -- 'class_teacher' | 'subject' | 'assistant'
)
students (
```

```
  id, school_id, branch_id, student_code, name_bn, name_en,
  roll, section_id, admission_date, date_of_birth, gender,
  blood_group, religion, nationality, nid_birth_cert,
  photo_url, guardian_phone, address_present, address_permanent,
  previous_school, status                    -- active | transferred | passed_out | dropped
)
student_guardians (
  id, student_id, user_id, relation,
  is_primary, can_pay_fees BOOLEAN, occupation, monthly_income
)
-- NEW: admission inquiry / lead funnel
admission_inquiries (
  id, school_id, branch_id, student_name, guardian_name, guardian_phone,
  class_interested, source,                  -- walk_in | phone | online | referral
  status,                                    -- new | contacted | visited | admitted | lost
  followup_date, assigned_to, notes, created_at
)
-- NEW: student transfer history
student_shifts (
  id, student_id, from_section_id, to_section_id,
  from_academic_year, to_academic_year,
  reason, shifted_by, shifted_at
```

```
)
```

5.3 Attendance

```
attendance (

  id, school_id, student_id, section_id, date,

  status,                          -- present | absent | late | leave | holiday

  check_in_time, check_out_time,      -- for biometric/QR

  marked_by, marked_at, remarks,

  UNIQUE(student_id, date)

)

attendance_devices (id, school_id, branch_id, name, type, device_id)

  -- type: biometric | rfid | qr_scanner
```

5.4 Exams & Marks

```
exams (id, school_id, academic_year_id, name, type, start_date, end_date)

exam_subjects (id, exam_id, subject_id, section_id, date, start_time,

  duration_mins, full_marks, pass_marks)

-- NEW: seating arrangement

exam_rooms (id, school_id, branch_id, name, capacity, rows, cols)

exam_seating (id, exam_subject_id, room_id, student_id, seat_row, seat_col)

marks (

  id, school_id, exam_subject_id, student_id,

  marks_obtained, grade, is_absent,

  entered_by, entered_at, locked BOOLEAN, approved_by, approved_at
```

```
)

grading_scales (id, school_id, name, rules JSONB)

-- Report cards with AI-generated comments

report_cards (

  id, school_id, student_id, exam_id,

  overall_gpa, position_in_class, attendance_pct,

  teacher_comment, ai_generated_comment, principal_remark,

  pdf_url, published_at, published_by

)
```

5.5 Fees & Payments

```
-- 27 fee heads seeded by default

fee_heads (

  id, school_id, name_bn, name_en, type, default_amount,

  applies_to,                          -- all | class | section | stream

  is_recurring BOOLEAN, frequency              -- monthly | quarterly | annual | one_time

)

fee_structures (id, school_id, class_id, fee_head_id, amount, frequency)

fee_invoices (

  id, school_id, student_id, month, year, total_amount,

  paid_amount, due_amount, status, due_date, late_fee,

  invoice_no UNIQUE
```

```
)

fee_invoice_items (id, invoice_id, fee_head_id, amount, waiver, waiver_reason)

payments (

  id, school_id, invoice_id, amount, method,

  -- method: cash | bkash | nagad | rocket | card | bank_transfer | cheque

  transaction_id, gateway_ref, paid_at, received_by, status,

  receipt_no UNIQUE, refund_status

)

-- NEW: per-student ledger (derived; snapshot for quick view)

student_ledger_entries (

  id, school_id, student_id, date, ref_type, ref_id,

  debit, credit, running_balance, note

)

-- NEW: scholarships

scholarships (id, school_id, name, criteria JSONB, amount_type, amount)

student_scholarships (id, student_id, scholarship_id, granted_by, valid_until)
```

5.6 Expenses, Income, Investments, Donations

```
-- 24 expense heads seeded by default

expense_heads (id, school_id, name_bn, name_en, category)

expenses (

  id, school_id, branch_id, date, head_id, amount,

  paid_to, payment_method, reference_no, description,
```

```
  approved_by, attachment_url, created_by
)
-- Cash receive types (admin receive, bank withdraw, loan, etc.)
cash_receive_heads (id, school_id, name_bn, name_en)
cash_receives (
  id, school_id, branch_id, date, head_id, amount,
  received_from, method, reference_no, received_by
)
-- NEW: investments
investments (
  id, school_id, title, principal, return_expected,
  start_date, maturity_date, status, notes
)
investment_returns (id, investment_id, date, amount, note)
-- NEW: donations (চাঁদা) — critical for madrasas
donation_campaigns (
  id, school_id, title, description, target_amount,
  start_date, end_date, status
)
donations (
  id, school_id, campaign_id, donor_name, donor_phone,
```

```
  donor_address, amount, method, received_by, received_at,

  receipt_no UNIQUE, is_anonymous

)

-- Salary / payroll (basic v1)

staff_salaries (

  id, school_user_id, month, year, basic, allowances JSONB,

  deductions JSONB, net_amount, status, paid_on, paid_by

)
```

5.7 Communication

```
notices (

  id, school_id, branch_id, title, body, audience,

  target_ids JSONB, attachments JSONB, sent_at, created_by,

  channels JSONB                          -- ['inapp','sms','whatsapp','push','email']

)

-- Unified messaging queue

message_queue (

  id, school_id, channel, recipient, body, status,

  provider, cost, scheduled_for, sent_at, error

)

sms_logs        (id, school_id, recipient, message, status, provider, cost, sent_at)

whatsapp_logs   (id, school_id, recipient, template_name, variables JSONB, status, provider,
cost, sent_at)
```

push_logs (id, school_id, user_id, title, body, status, sent_at)

conversations (id, school_id, subject, created_by, allow_parent_teacher BOOLEAN)

messages (id, conversation_id, sender_id, body, attachments JSONB, sent_at, read_by JSONB)

5.8 Certificates

```
certificate_templates (

  id, school_id, type,                -- testimonial | tc | character | completion | hifz_sanad

  name, html_template, variables JSONB, orientation, paper_size

)

certificates_issued (

  id, school_id, student_id, template_id,

  issued_on, serial_no UNIQUE, issued_by, pdf_url, signed_qr

)
```

5.9 Support Tickets

```
support_tickets (

  id, school_id, created_by, subject, body, priority,

  status, assigned_to, category, created_at

)

support_messages (id, ticket_id, user_id, body, attachments JSONB, created_at)
```

5.10 Madrasa-Specific (NEW — feature-flagged for madrasa tenants)

```
-- Hifz progress (Quran memorization)

hifz_progress (

  id, school_id, student_id,
```

```
  para_no,                          -- 1..30
  sipara_no,                        -- pages within para
  status,                           -- learning | revising | completed | tested
  tested_on, tested_by, mark, grade,
  mistakes_count, note
)
-- Kitab curriculum (Ibtedaiyyah → Mutawassita → Sanaweeya → Fazilat → Kamil)
kitab_curriculum (id, school_id, class_id, stage, kitab_name, author, display_order)
student_kitab_progress (
  id, student_id, kitab_id, started_on, current_chapter,
  completed_on, grade, teacher_id
)
-- Daily Sabaq / Sabqi / Manzil log (core madrasa daily record)
daily_sabaq (
  id, school_id, student_id, date,
  sabaq_para, sabaq_from, sabaq_to, sabaq_quality,    -- excellent | good | average | weak
  sabqi_para, sabqi_quality,
  manzil_paras JSONB, manzil_quality,
  tajweed_notes, teacher_id, UNIQUE(student_id, date)
)
```

5.11 Library, Transport, Hostel, Canteen, Inventory, Health

-- Library

library_books (id, school_id, title, author, isbn, category, copies_total, copies_available, shelf)

library_issues (id, book_id, student_id, issued_on, due_on, returned_on, fine)

-- Transport (with GPS — v2)

transport_routes (id, school_id, name, start_point, end_point, fare_per_month)

transport_vehicles (id, route_id, reg_no, driver_name, driver_phone, capacity, gps_device_id)

transport_students (id, student_id, route_id, vehicle_id, pickup_stop, pickup_time)

transport_gps_pings (id, vehicle_id, lat, lng, speed, recorded_at) -- TimescaleDB / realtime

-- Hostel

hostels (id, school_id, branch_id, name, gender, capacity)

hostel_rooms (id, hostel_id, room_no, capacity)

hostel_allocations (id, room_id, student_id, from_date, to_date)

-- Canteen (prepaid wallet)

canteen_wallet (id, student_id, balance, updated_at)

canteen_transactions (id, student_id, amount, type, item_name, merchant_id, at)

-- Inventory (uniforms, books, stationery)

inventory_items (id, school_id, name, category, unit, stock, reorder_level, unit_price)

inventory_movements (id, item_id, type, qty, reference, date, by_user)

-- Health / medical

student_health (

```
  id, student_id, blood_group, allergies, chronic_conditions,

  medications, emergency_contact, doctor_name, doctor_phone,

  vaccinations JSONB, last_checkup_date

)

-- Counseling / disciplinary (principal + counselor only)

counseling_logs (

  id, student_id, date, type, summary, action_taken,

  confidential BOOLEAN, created_by

)
```

5.12 LMS (v2 — Phase 4+)

assignments (id, school_id, subject_id, section_id, title, description, due_date, max_marks)

assignment_submissions (id, assignment_id, student_id, submitted_at, file_url, marks, feedback)

online_classes (id, subject_id, section_id, scheduled_at, duration, meet_url, provider, recording_url)

5.13 Public School Website per tenant (v2)

public_pages (id, school_id, slug, title, body_html, seo_meta JSONB, published)

public_gallery (id, school_id, album, image_url, caption, display_order)

public_news (id, school_id, title, body, image_url, published_at)

---

6. Default Seed Data (must ship on tenant creation)

6.1 Fee Heads (27 — matches Bornomala + extras)

Tuition Fee, Tuition Fee Due, Admission Form, Admission Fee, Admission Fee Due, Session Fee, Session Fee Due, Registration Fee, Package Fee, Exam Fee — 1st Term, Exam Fee — 2nd Term, Exam Fee — 3rd Term, Exam Fee — Final, Transport Fee, Fine, ID Card Fee, Online Charge, Food Expense, Testimonial Fee, Sports Fee, Computer Lab Fee, Practical Fee, TC Fee, Syllabus Fee, Certificate Fee, Books Fee, Note Book Fee, Stationeries Fee, Development Fee, Dress Fee.

6.2 Expense Heads (24)

Admin Personal, Advance Payment, Asset, Bank Deposit, Bonus Pay, Voucher Bill, Electricity Bill, Entertainment, General Purchase, Honorarium Pay, Internet Bill, Investment Return, Labour Bill, Loan Paid, Office Expense, Office Rent, Overtime Bill, Product Purchase, Salary Pay, Software Bill, Stationery, Total Salary, Transport, Others.

6.3 Cash Receive Heads (8)

Admin Receive, Boss Personal, Bank Withdraw, Loan Receive, Loan Return, Opening Balance, Project Investment, Others.

6.4 Grading Scale (Bangladesh standard GPA 5.0)

A+ (80-100, 5.00) · A (70-79, 4.00) · A- (60-69, 3.50) · B (50-59, 3.00) · C (40-49, 2.00) · D (33-39, 1.00) · F (<33, 0.00).

6.5 Madrasa tenants — auto-seed additionally

- Kitab stages: Ibtedaiyyah, Mutawassita, Sanaweeya Aamma, Sanaweeya Khassa, Fazilat, Kamil.
- Hifz paras 1–30.
- Sabaq quality scale: Excellent / Good / Average / Weak.

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7. Integrations (Bangladesh)

7.1 Payments (priority order)

1. **SSLCommerz** (covers cards + bKash + Nagad + Rocket + bank) — ship first.
2. **bKash Tokenized Checkout** direct.

3. **Nagad Payment Gateway** direct.
4. **Bank transfer** manual reconciliation.

7.2 Messaging

1. **SMS:** SSL Wireless, Alpha-SMS, BulkSMSBD (per-tenant credits).
2. **WhatsApp Business API:** via Interakt / Twilio (template messages for attendance, fees, results).
3. **Push notifications:** Firebase Cloud Messaging (PWA + future native).
4. **Email:** Resend / AWS SES.

7.3 Biometric / RFID / QR

- **ZKTeco biometric device API** for attendance punch.
- **QR attendance:** student QR → gate scanner → instant SMS to parent.
- **RFID card gate entry.**

7.4 Calendars & Fonts

- Hijri calendar (moment-hijri) + Bengali calendar alongside Gregorian.
- Unicode fonts: Noto Sans Bengali, Hind Siliguri, Kalpurush. **No SutonnyMJ / Bijoy.** Arabic: Noto Naskh Arabic.

7.5 AI (Anthropic / OpenAI)

- Report card comment generator.
- Dropout risk prediction (attendance trend + marks trend).
- Smart SMS composer (suggest message text).
- Fee reminder tone optimizer.

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8. Next.js Folder Structure

```
src/

├── app/

│   ├── (marketing)/                # Public landing, pricing

│   │   ├── page.tsx

│   │   └── pricing/page.tsx
```

```
│   ├── (auth)/
│   │   ├── login/page.tsx
│   │   ├── register-school/page.tsx
│   │   └── forgot-password/page.tsx
│   ├── super-admin/                # Dashboard 1
│   │   ├── layout.tsx
│   │   ├── page.tsx
│   │   ├── schools/
│   │   ├── subscriptions/
│   │   ├── plans/
│   │   ├── feature-flags/
│   │   └── analytics/
│   └── school/[schoolSlug]/
│       ├── layout.tsx              # tenant context + membership check + branch switcher
│       ├── admin/                  # Dashboard 2 (Principal)
│       │   ├── layout.tsx
│       │   ├── page.tsx            # live dashboard with charts
│       │   ├── admission-inquiry/      # NEW
│       │   ├── staff/
│       │   │   └── [id]/permissions/
│       │   ├── students/
│       │   │   ├── new/
```

```
│   │   │   ├── bulk-import/
│   │   │   ├── [id]/
│   │   │   │   ├── page.tsx
│   │   │   │   ├── ledger/        # NEW per-student ledger
│   │   │   │   └── edit/
│   │   │   └── shift/             # NEW transfer workflow
│   │   ├── classes/
│   │   ├── sections/
│   │   ├── subjects/
│   │   ├── attendance/
│   │   ├── exams/
│   │   │   ├── routine/
│   │   │   ├── admit-card/
│   │   │   ├── seating/           # NEW
│   │   │   ├── marks-entry/
│   │   │   └── results/
│   │   ├── certificates/          # NEW
│   │   ├── fees/
│   │   │   ├── heads/
│   │   │   ├── structures/
│   │   │   ├── invoices/
```

```
│   │   │   └── payments/
│   │   ├── expenses/
│   │   ├── income/
│   │   ├── investments/          # NEW
│   │   ├── donations/            # NEW (campaigns + receive)
│   │   ├── payroll/
│   │   ├── notices/
│   │   ├── messaging/            # SMS + WhatsApp + Push composer
│   │   ├── reports/              # 20+ report types
│   │   ├── library/
│   │   ├── transport/
│   │   ├── hostel/
│   │   ├── inventory/
│   │   ├── support/              # NEW tickets
│   │   ├── branches/             # NEW multi-branch
│   │   ├── madrasa/              # NEW — feature-flagged
│   │   │   ├── hifz/
│   │   │   ├── kitab/
│   │   │   └── daily-sabaq/
│   │   └── settings/
│   ├── teacher/                  # Dashboard 3
│   │   ├── layout.tsx
```

```
│   │   ├── page.tsx
│   │   ├── attendance/          # offline-capable
│   │   ├── marks/               # offline-capable
│   │   ├── assignments/
│   │   ├── daily-sabaq/         # for madrasa ustadh
│   │   └── messages/
│   ├── portal/                  # Dashboard 4 (parent/student)
│   │   ├── layout.tsx
│   │   ├── page.tsx
│   │   ├── children/            # multi-child switcher (parent)
│   │   ├── attendance/
│   │   ├── results/
│   │   ├── fees/
│   │   │   └── pay/             # bKash/Nagad/Rocket/Card flow
│   │   ├── assignments/
│   │   ├── hifz/                # student's own progress
│   │   ├── notices/
│   │   └── messages/
│   └── public/                  # v2 — public school website
│       ├── page.tsx             # about
│       ├── admission/
```

```
│           ├── notices/
│           └── gallery/
├── components/
│   ├── ui/
│   ├── charts/                    # Recharts wrappers
│   ├── forms/
│   ├── tables/
│   └── offline/                   # IndexedDB sync components
├── lib/
│   ├── supabase/
│   │   ├── client.ts
│   │   ├── server.ts
│   │   └── admin.ts
│   ├── auth/
│   │   ├── permissions.ts
│   │   └── middleware.ts
│   ├── payments/
│   │   ├── sslcommerz.ts
│   │   ├── bkash.ts
│   │   └── nagad.ts
│   ├── messaging/
│   │   ├── sms.ts
```

```
│   │   ├── whatsapp.ts
│   │   └── push.ts
│   ├── pdf/                        # react-pdf templates
│   │   ├── marksheet.tsx
│   │   ├── admit-card.tsx
│   │   ├── receipt.tsx
│   │   ├── certificate.tsx
│   │   └── hifz-sanad.tsx          # madrasa
│   ├── ai/                         # Claude integration
│   │   ├── report-comment.ts
│   │   └── dropout-risk.ts
│   ├── i18n/                       # bn / en / ar
│   ├── calendars/                  # bengali + hijri
│   └── utils/
├── hooks/
│   ├── usePermission.ts
│   ├── useActiveSchool.ts
│   ├── useActiveBranch.ts
│   ├── useRealtime.ts
│   └── useOfflineQueue.ts
├── server/
```

```
│   └── actions/
│       ├── admission-inquiry.ts       # NEW
│       ├── students.ts
│       ├── student-ledger.ts          # NEW
│       ├── student-shift.ts           # NEW
│       ├── attendance.ts
│       ├── marks.ts
│       ├── fees.ts
│       ├── payments.ts
│       ├── donations.ts               # NEW
│       ├── investments.ts             # NEW
│       ├── hifz.ts                    # NEW
│       ├── daily-sabaq.ts             # NEW
│       └── certificates.ts            # NEW
├── middleware.ts                      # route protection + tenant + branch resolution
├── service-worker.ts                  # PWA + offline
└── manifest.ts                        # PWA manifest
```

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9. Reports (20 types — exceeds Bornomala's 13)

Financial (11): Fee Collection, Fee Item, Tuition Fee Summary, Tuition Due, Fee Total Summary, Admission + Session Fee, Donation Report, Expense Report, Cash Sheet, Cash Receive Summary, Received & Payment Summary, Investment Return.

Academic (5): Attendance Trend (30/90 day), Exam Performance (class-wise + subject-wise), Teacher Performance, Dropout Risk (AI), Enrollment Funnel (lead → admitted).

Madrasa (2): Daily Sabaq Report, Hifz Progress Report.

Operational (4): Dues Aging (0-30/31-60/61-90/90+), SMS/WhatsApp Usage, Audit Log, Branch Comparison.

Each report: filter by date range, branch, class, section, student. Export: PDF (Bangla Unicode) + Excel.

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10. Subscription Plans

Plan	Price/mo (BDT)	Students	Branches	SMS	Storage	Features
Free Trial	0 (30 days)	50	1	100	500 MB	Core
Basic	1,500	300	1	1,000	5 GB	Core + SMS + Parent portal
Pro	3,500	1,000	2	5,000	20 GB	+ Online payments + WhatsApp + AI reports
Madrasa Pro	3,500	1,000	2	5,000	20 GB	Pro + Hifz + Kitab + Sabaq + Hijri
Enterprise	Custom	Unlimited	Unlimited	Custom	Custom	All + Biometric + GPS + Custom domain + API + SLA

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11. Implementation Phases

Phase 0 — Foundation (Week 1–2)

- Repo, CI/CD, Supabase project, env setup
- Auth (email + phone OTP + magic link)
- Tenant + branch resolution middleware
- Base layouts for all four dashboards
- RLS scaffolding
- Default seed: 27 fee heads, 24 expense heads, 8 cash-receive heads, GPA 5.0 scale

Phase 1 — Core Academic MVP (Week 3–6)

- School settings, branches, classes, sections, subjects
- **Admission Inquiry pipeline** (NEW)
- Student admission (single + Excel bulk + **shift/transfer** NEW)
- **Per-student Ledger** (NEW)
- Staff management + **granular permission UI**
- Class + Subject teacher + Madrasa Ustadh assignment
- Attendance (daily, class-wise)
- Parent/Student portal (read-only)
- **PWA scaffold** (install, offline shell)

Phase 2 — Exams, Results, Certificates (Week 7–9)

- Exam setup, grading scales
- **Seating arrangement** (NEW)
- Marks entry (scoped permission)
- Result publishing workflow
- Marksheet PDF (Bangla Unicode)
- **AI-generated report card comments** (NEW)
- **Certificates** (testimonial, TC, character, completion, **Hifz Sanad**) (NEW)
- Parent result view

Phase 3 — Fees, Payments, Donations, Investments (Week 10–12)

- Fee heads, structures, invoice generation (monthly cron)
- Cash collection flow
- **SSLCommerz** integration (covers cards + MFS)
- **bKash + Nagad** direct
- Receipts, dues report, collection dashboard

- **Donation campaigns + receive** (NEW — madrasa critical)
- **Investment tracking + returns** (NEW)
- Payroll basic
- **Scholarship workflow** (NEW)

Phase 4 — Communication (Week 13–14)

- Unified messaging: SMS + WhatsApp Business + Push + In-app + Email
- Notice composer (multi-channel)
- Admit card generator
- Teacher ↔ Parent messaging (principal toggle)
- SMS/WhatsApp templates + usage report
- **Support ticket system** (NEW)

Phase 5 — Madrasa Module (Week 15–16) — feature flag

- **Hifz progress tracker** (para/sipara, daily tested)
- **Kitab curriculum** (stages + books + student progress)
- **Daily Sabaq / Sabqi / Manzil log**
- **Hifz Sanad certificate**
- **Hijri calendar** + Arabic UI support (RTL)

Phase 6 — Offline, Live, Scale (Week 17–18)

- PWA full offline: attendance + marks entry (IndexedDB + sync)
- Realtime: live attendance dashboard, notice broadcast, messages (Supabase Realtime)
- Super Admin dashboard + billing + feature flags
- **Multi-branch switcher + consolidated reports**
- Library, Transport (+ GPS v2), Hostel, Canteen, Inventory
- 20 report types fully wired

Phase 7 — AI & Advanced (Week 19–20)

- Dropout risk ML model
- Smart SMS composer
- Audit log viewer
- 2FA, session management
- Automated backup + data export
- Webhook/API for govt reporting (EIIN, EMIS)
- Command palette (⌘K), keyboard shortcuts

Phase 8 — Platform (v2, Week 21+)

- Public school website per tenant
- Custom domain (Enterprise)

- LMS: homework submission, online classes (Jitsi/Meet)
- e-Library (digital books, past papers)
- PTA community, Alumni tracking
- Marketplace (plugins by third parties)
- Tri-lingual (bn/en/ar)
- Native mobile apps (React Native on top of shared logic)

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12. 10 Tier-1 Killer Differentiators (ship in Phase 0–5)

1. **Online fee payment** — bKash/Nagad/Rocket/SSLCommerz (Phase 3).
2. **Parent mobile PWA** — app-like, offline-capable (Phase 1–6).
3. **Student portal** — own view, attendance streaks, homework feed (Phase 1).
4. **Live realtime dashboard** — attendance ring updates instantly (Phase 0–6).
5. **Granular scoped permissions** — class/subject/student scope (Phase 1).
6. **Offline attendance + marks entry** — IndexedDB sync (Phase 6).
7. **WhatsApp + SMS + Push + Email** — unified (Phase 4).
8. **QR attendance + ID card** — instant parent SMS (Phase 6).
9. **Madrasa module** — Hifz/Kitab/Sabaq (Phase 5).
10. **Multi-branch / multi-campus** — one login, N branches (Phase 6).

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13. Security Checklist

- ☐ RLS enabled on EVERY tenant table
- ☐ Service-role key only in server-side Super Admin context
- ☐ All server actions call `requirePermission()`
- ☐ JWT contains `active_school_id` + `active_branch_id` claims
- ☐ Rate limit login, OTP, payment endpoints
- ☐ Audit log on all mutations to student/marks/fee/user_permissions/donations
- ☐ Daily Supabase backups + weekly offsite
- ☐ PII encrypted at rest; no PII in logs
- ☐ CSRF protection on forms
- ☐ File uploads scanned (Edge Function + ClamAV or tier service)
- ☐ 2FA for Principal and Super Admin roles
- ☐ Device session management + revoke

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14. Testing Strategy

- Unit: Vitest for permission helpers, fee calc, grading
- Integration: Playwright for each dashboard flow
- **Permission matrix test** — auto-test every role × every endpoint (positive + negative)
- Load test: k6 at 9am peak attendance (500 teachers simultaneously)
- Accessibility: axe-core on every page
- i18n snapshot: render all pages in bn / en / ar

---

15. First Concrete Tasks for Claude Code

1. Init Next.js 14 + TypeScript + Tailwind + shadcn/ui + next-intl + PWA.
2. Supabase local dev (`supabase init`, `supabase start`).
3. Migrations for all tables in §5.1–§5.11 with RLS.
4. Seed script for §6 defaults (fee heads, expense heads, cash receive heads, grading scale; conditional madrasa seeds).
5. `lib/supabase/server.ts`, `client.ts`, `admin.ts`.
6. Auth: `/login`, `/register-school` (creates schools + school_branches primary + school_users row + active_school_id claim).
7. `middleware.ts`: resolve schoolSlug + active branch + role.
8. Four dashboard route groups with role-gated layouts.
9. `requirePermission()` helper + `usePermission()` hook.
10. **Principal's Manage Staff page with permission grant UI** — checkboxes per action × resource, scoped by class/section/subject.
11. **Admission Inquiry pipeline** UI + data.
12. Class + Section + Subject CRUD.
13. Student admission (single + xlsx bulk + shift/transfer).
14. **Per-student Ledger** page.
15. **Attendance module end-to-end** (teacher entry + parent view) — first vertical slice exercising the permission system + realtime.

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16. Non-Goals (v1)

- Native mobile apps (PWA suffices for v1; RN in v2).
- Advanced accounting (double-entry ledger, balance sheet) — basic cash-in/out only.
- Video classes (integrate Jitsi/Meet in v2).
- Custom domain per school (Enterprise v2).

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17. Open Questions for Owner

1. Path-based tenant (`/school/slug/...`) vs subdomain? → **Recommend path-based v1.**
2. SMS provider default? → need credentials for testing.
3. First payment gateway? → **Recommend SSLCommerz first.**
4. Madrasa module — enable by default for type='madrasa' tenants? → **Yes.**
5. Default grading? → **GPA 5.0 standard.**
6. AI report card comments — enable per plan or as add-on? → Tier-aware.

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v2 Change-log vs v1

- Added: admission_inquiries, student_shifts, student_ledger_entries, donation_campaigns + donations, investments, exam_rooms + exam_seating, certificate_templates + certificates_issued, support_tickets, school_branches, madrasa tables (hifz_progress, kitab_curriculum, daily_sabaq), transport GPS tables, canteen wallet, inventory, health & counseling, scholarships, payroll, public website.
- Added: 27 fee heads + 24 expense heads + 8 cash receive heads default seeds.
- Added: WhatsApp + Push + Email channels (not just SMS).
- Added: SSLCommerz / bKash / Nagad payment integration.
- Added: PWA + offline sync for attendance + marks.
- Added: Madrasa module (Phase 5) as feature flag.
- Added: Multi-branch support.
- Added: AI report card comments + dropout risk prediction.
- Added: 20 report types (was 8).
- Added: Super Admin feature flags, support ticket system.
- Added: Hijri + Bengali calendars, tri-lingual support (bn/en/ar).
- Added: Madrasa-specific roles (MADRASA_USTADH), BRANCH_ADMIN role.
- Changed: Folder structure expanded with new routes for all additions.
- Changed: Phases restructured from 5 to 8 (Phase 5 = Madrasa, Phase 7 = AI, Phase 8 = Platform).

End of Plan v2. Hand this file to Claude Code and ask it to begin with Phase 0 §15.